

Large-Scale Photonic Ising Machine by Spatial Light Modulation

slleung

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In this Letter, we propose and experimentally demonstrate the use of spatial light modulation for calculating the ground state of an Ising Hamiltonian. The phase matrix on a spatial light modulator (SLM) acts as a lattice of spins whose interaction is ruled by the constrained optical intensity in the far-field and can be programmed by input amplitude modulation. Feedback from the detection plane allows the spatial phase distribution to evolve towards the minimum of the selected spin model. We find ferromagnetic-like ground-states in agreement with mean-field predictions. Our spatial Ising machine hosts thousands of parallelly-processed spins, and represents a scalable and efficient approach for photonic computing. We implement a spatial photonic Ising machine by using the phases in separated spatial points of the optical wavefront. A binary phase modulated beam encodes binary spins with configurable interactions [Fig. 1(a)]. A spin variable $\sigma_j = e^{i\phi_j} = \pm 1$ corresponds to a spatial point of the optical field with phase $\phi_j \in \{0, \pi\}$. As illustrated in Fig. 1(b-c), an SLM acting as a reprogrammable matrix of pixels imprints binary phase values on the coherent wavefront. Setting the SLM in the Fourier space of the electric field $\tilde{E}(k)$, we have

$$\tilde{E}(k) = \sum_j \xi_j \sigma_j \tilde{\delta}_W(k - k_j) \quad (1)$$

where ξ_j indicates the field amplitude incoming on each pixel. The normalized rectangular function $\tilde{\delta}_W$ models the pixel of finite size $2W$, so that $k_j = 2Wj$, with $j = 1; \dots, n$. The resulting far-field intensity after free-space propagation is

$$I(x) = |E(x)|^2 = \sum_{jh} \xi_j \xi_h \sigma_j \sigma_h \delta_W^2(x) e^{2iW(h-j)x} \quad (2)$$

with $\delta_W(x) = \sin(Wx)/(Wx)$ the inverse Fourier transform of $\tilde{W}(k)$. Spin-spin interaction can be induced by acting on the intensity on the detection plane. We constrain $I(x)$ by a measurement and feedback method to couple the phases on the SLM plane. Minimizing $\|I_T(x) - I(x)\|$ for an arbitrary target intensity image $I_T(x)$ thus corresponds to minimizing a Hamiltonian H .

After normalization, $\int [I_T(x)]^2 dx \simeq \int [I(x)]^2 dx$, and the function H takes the form of the Ising Hamiltonian

$$H = - \sum_{jh} J_{jh} \sigma_j \sigma_h \quad (3)$$

with spin interactions given by

$$J_{jh} = 2\xi_j \xi_h \int I_T(x) \delta_W^2(x) e^{2iW(h-j)x} dx \quad (4)$$

When the effect of the SLM pixel size can be neglected, $\delta_W(x) \rightarrow 1$, and the couplings reduce to

$$J_{jh} = 2\pi \xi_j \xi_h \tilde{I}_T[2W(j-h)] \quad (5)$$

which indicates that the interaction matrix is set by the input amplitude modulation along with the Fourier transform of the far-field target image. The interaction passes from short- to long-range by changing the spatial profile of I_T .

In particular, in the case of a point-like target image, the spins are all-to-all interacting ($J_{jh} = \text{const}$) for an input wave with constant amplitude. Using a programmable (quenched) amplitude mask on the input beam the couplings can be varied according to $J_{jh} \propto \xi_j \xi_h$, which allows to implement the entire class of spin-glass models, known as Mattis models [32, 33], where the pairwise interaction can be expressed as product of two independent variables. Fig. 1(c) shows the principle of operation of our Ising machine. A spin configuration σ_j is generated upon an amplitude-modulated wavefront using binary phases on the SLM and the corresponding intensity distribution $I(x)$ is measured in the far-field. The detected image is compared with the target $I_T(x)$ and the information is feedback to the SLM plane. The system evolves towards minimization of the cost function $f = ||I_T(x) - I(x)||$, which corresponds to the Ising ground state.

The experimental optical machine follows the setting shown in Fig. 1(b). Light from a continuous-wave laser source with wavelength $\lambda = 532\text{nm}$ is expanded, eventually modulated in amplitude, and impinges on a nematic liquid crystal reective modulator (Holoeye LC-R 720, 1280×768 pixels, pixel pitch $20 \times 20\mu\text{m}$) whose active area is selected by a rectangular aperture to host $N = L \times L$ spins (pixels). The SLM is set into a phase-modulation mode with less than 10% residual intensity modulation by a combination of incident and analyzed polarizations. Phase-modulated light is spatially filtered (3mW power) and then focused by a lens ($f = 500\text{mm}$) on a CCD camera. The intensity is measured on $M = 18 \times 18$ spatial modes obtained grouping 16×16 camera pixels to average over speckles arising from spatial phase fluctuations in the far-field plane.

We first demonstrate the spatial Ising machine for $N = 4 \times 104$ spins with all-to-all couplings ($J_{jh} = \text{const}$), which corresponds to a number of spin-spin connections orders of magnitude larger than those realized in timemultiplexed platforms [19, 20]. In this case, $\xi_j = \xi_h = \xi_0$ and the target corresponds to intensity focused only in a single spatial mode, that is, a bright localized spot

[Fig. 2(a)]. The binary phases on the SLM are initialized by a random distribution, which gives a weak and broad speckle pattern in the detection plane. By a Monte Carlo-like method, at each iteration we randomly flip a small cluster of spins and measure the corresponding far-field intensity, retaining the change only if its difference with the target image decreases [34]. Unlike other photonic Ising machines [23], no information about the target Hamiltonian is used to affect electronically the spin evolution. To prevent trapping into local minima induced by the algorithm, we select clusters with a gradually increasing size. To follow the system evolution we consider as physical observables the energy H and the magnetization $m = \langle \sigma_j \rangle$ of each configuration. As shown in Fig. 2(b) for different realizations, we observe a monotonic growth of $|m|$, which saturates to a large value after approximately 103 iterations. The Hamiltonian monotonically decreases toward a plateau, thus indicating the onset of a low-energy ferromagnetic-like state. The actual temperature T of these spin configurations is determined by the random phase fluctuations in the Fourier plane, which results from the intrinsic noise characterizing each operation in the experimental setup. Sources of noise come from the quantization on the CCD discrete modes of the detected intensity as well as from the imperfect spatial phase modulation [35].

1 Summary

Spin is a fundamental property of particles, including photons (particles of light). It's a form of angular momentum, like a spinning top. For photons, spin is related to their polarization: circularly polarized light carries spin angular momentum. However, in the context of the photonic Ising machine, spin is not directly related to the physical spin of photons. Instead, it's a computational concept where a spin can have two values: +1 or -1. These values are mapped to physical properties of light, such as phase or intensity.

The experimental setup is relatively straightforward compared to other photonic computing approaches because it primarily relies on well-established optical components like spatial light modulators and detectors. The core idea of encoding information in light's phase and using optical interference for calculations is conceptually simple.

Steps Involved in the paper:

1. Encoding spin variables: Convert binary spin values (+1, -1) into phase modulations of a light field using a spatial light modulator.
2. Light propagation: The modulated light propagates through an optical system. Interactions between different light paths create interference patterns.
3. Energy calculation: The intensity of the interference pattern is related to the energy of the Ising system.

4. Optimization: The system iteratively adjusts the phase modulations to minimize the energy, corresponding to finding the ground state of the Ising Hamiltonian.
5. Decoding: The final phase modulations are converted back into binary spin values, representing the solution to the optimization problem.

Minimizing energy corresponds to finding the ground state. The ground state of an Ising system is the configuration with the lowest energy. In the photonic Ising machine, this corresponds to the light intensity pattern with the minimum total intensity. The system iteratively adjusts the phase modulations to reduce the total intensity, eventually converging to the ground state.

The phase of light represents spin values. Spin +1 represent to the phase shift of 0 degrees, while spin -1: Phase shift of 180 degrees. The spatial light modulator (SLM) can create a complex pattern of phase shifts, representing a combination of spin variables.

The light propagation through the optical system is designed to mimic the interactions between spins in the Ising model. The interference patterns that form represent the energy of the system. The goal is to find the phase configuration that minimizes this energy, which corresponds to the ground state of the Ising Hamiltonian.

General Questions:

Scalability: What are the limitations on the number of spins that can be simulated using this approach? How does the performance scale with the system size? Noise and Error: How do noise and errors in the system affect the accuracy of the results? What techniques can be used to mitigate these effects? Comparison with Other Platforms: How does this photonic Ising machine compare to other implementations, such as superconducting qubit-based or trapped-ion-based systems? Applications: What are the potential applications of this photonic Ising machine beyond the examples discussed in the paper? Specific Questions Related to the Paper:

Mattis Model: How does the performance of the machine vary for different types of spin-glass models, such as non-Mattis models? Hamiltonian Optimization: Can the machine be used to solve other types of optimization problems beyond the Ising model? Quantum Advantage: Is there a potential for this photonic Ising machine to demonstrate a quantum advantage over classical computers for certain problem instances? Experimental Challenges: What are the main experimental challenges in implementing and operating this photonic Ising machine? How can these challenges be addressed?

2 Introduction

Complex systems, such as those found in biology and social science, are characterized by a vast number of possible internal states. This exponential growth in states with system size makes them extremely difficult to simulate or optimize using traditional computers. A key challenge in studying complex systems is their state space. This refers to the set of all possible configurations of the system. For even moderately sized systems, the state space grows exponentially, making it computationally intractable to explore all possibilities. This phenomenon is often referred to as the curse of dimensionality. Optimization problems arise when seeking the best possible solution from a vast number of alternatives. In the context of complex systems, these problems are often computationally demanding and belong to the class of NP-hard problems. This means there's no known efficient algorithm to find the optimal solution in polynomial time.

Researchers are exploring various physical systems, including those based on quantum and classical physics, to implement Ising models. While optical systems offer the potential for a large number of "spins," controlling these spins and their interactions remains a significant challenge.

The prospect of harnessing light for computational tasks has garnered significant attention due to the inherent advantages of optical systems: high speed and inherent parallelism. A particularly promising avenue within this domain is the development of photonic platforms designed to tackle complex computational problems.

One such class of problems is challenging for traditional computers, which involves complex systems characterized by a vast number of interacting components. These systems, ubiquitous in fields from biology to social sciences, often defy efficient computational analysis due to the exponential growth of their potential states. A computational framework that has shown promise in addressing these complexities is the Ising model, a physical model centered around interacting spins. By mapping complex problems onto the Ising model, researchers have sought to leverage the tools of physics to find optimal solutions.

In the realm of optics, various approaches have been explored to construct physical implementations of Ising machines. One method involves time-multiplexed optical parametric oscillators, where a series of optical pulses represent the spins in the Ising model. Another strategy entails the use of nanophotonic circuits, which integrate optical components on a chip to directly simulate spin interactions. Additionally, spatial light modulation has emerged as a potential technique for optical computing, although its application to Ising spin dynamics is still in its nascent stages.

While these approaches offer intriguing possibilities, challenges persist in

scaling up these systems to handle complex problems efficiently. Controlling a large number of optical elements to represent spins and engineering precise interactions between them remain significant hurdles. Moreover, accurately extracting the solution, or ground state, from the optical system presents another obstacle.

Despite these challenges, the potential of photonic platforms for revolutionizing computation is undeniable. As research progresses, we can anticipate advancements that will bring these concepts closer to practical applications.

2.1 Spatial Light Modulator (SLM)

The Spatial Light Modulator (SLM) can be used to implement a photonic Ising machine, where binary spin values are encoded in the phase of a light wave, which is a process of phase encoding. The spin σ_j at each spatial point of the wavefront corresponds to the phase ϕ_j in the electric field:

$$\sigma_j = \begin{cases} +1 & \text{if } \phi_j = 0, \\ -1 & \text{if } \phi_j = \pi \end{cases}, \quad j = 1, 2, 3, \dots, n.$$

ξ_j indicates the electric field amplitude incident on each pixel, which determines the strength of the electric contribution from the j -th spatial point.

$\delta_W(x)$ is a normalized sinc function with width $2W$ centered at $x = 0$, describing the spread in the spatial domain due to the band-limited nature of the Fourier components. It is used to approximate a rectangular function due to its smooth properties in the Fourier domain.

$$\delta_w(x) = \frac{\sin Wx}{Wx} \quad (6)$$

$\delta_W(x - x_j)$ represents the spatial distribution of the electric field contribution from the j -th spatial point, i.e., the spatial localization of the spins. It is centered at x_j and has a width determined by $2W$.

The electric field in the Fourier space $\tilde{E}(k)$ is expressed as a sum of contributions from each spatial frequency component k , modulated by the electric field amplitude ξ_j and phase ϕ_j .

$$\tilde{E}(k) = \sum_j \xi_j \sigma_j \tilde{\delta}_W(k - k_j) \quad (7)$$

$\tilde{\delta}_W(k - k_j)$ is the Fourier Transform of $\delta_w(x)$ in the frequency domain, representing a shifted rectangular function centered at k_j with a width of $2W$.

To find the spatial domain of the electric field $E(x)$, we perform the inverse Fourier Transform $E(k)$:

$$E(x) = \frac{1}{2\pi} \int \tilde{E}(k) e^{ikx} dk \quad (8)$$

Assuming that the sum and integral can be interchanged, substituting the expression and get

$$\begin{aligned} E(x) &= \frac{1}{2\pi} \int \sum_j \xi_j \sigma_j \tilde{\delta}_W(k - k_j) e^{ikx} dk \\ &= \sum_j \xi_j \sigma_j \frac{1}{2\pi} \int \tilde{\delta}_W(k - k_j) e^{ikx} dk \end{aligned} \quad (9)$$

where $\sum_j \xi_j \sigma_j e^{ik_j x}$ represents the superposition of waves with different spatial frequencies k_j , each modulated by the amplitude ξ_j and spin valuable σ_j .

The term $\frac{1}{2\pi} \int \tilde{\delta}_W(k - k_j) e^{ikx} dk$ is the inverse Fourier Transform of $\tilde{\delta}_W(k - k_j)$, which is a shifted sinc function and equal to

$$\frac{1}{2\pi} \int \tilde{\delta}_W(k - k_j) e^{ikx} dk = \delta_W(x - x_j) e^{ik_j x} \quad (10)$$

where $\delta_W(x)$ is the inverse Fourier Transform of $\tilde{\delta}_W(k)$. The function $\delta_W(x - x_j)$ represents the spatial distribution of the electric field centered at x_j , with the spread determined by δ_W . Therefore, the expression of $E(x)$ is

$$E(x) = \sum_j \xi_j \sigma_j \delta_W(x - x_j) e^{2ik_j x} = \delta_W(x - x_j) \sum_j \xi_j \sigma_j e^{2iWjx} \quad (11)$$

where $e^{2ik_j x}$ is a phase factor that includes the spatial frequency k_j , which is related to the pixel position on the SLM. The relationship $k_j = 2Wj$ maps the discrete spatial domain (pixels) to the continuous frequency domain. The spacing between adjacent $2Wj$ determines the frequency resolution of the system. A larger W results in a coarser frequency sampling. To avoid aliasing, the sampling frequency W should be at least twice the maximum frequency component present in the signal.¹

Therefore, $E(x)$ describes a spatial field distribution that is a product of a sinc function and a sum of modulated exponential terms, reflecting the wave-like nature of the phenomenon. The electric field is composed of multiple components, each centered at a different spatial point. In the Fourier domain, each

¹Consider a one-dimensional array of pixels. Each pixel contributes to a specific range of spatial frequencies. The equation $k_j = 2Wj$ maps the pixel index to the central frequency of this range. This equation provides a simplified model for relating the discrete spatial domain (pixels) to the continuous frequency domain.

pixel contributes to the overall frequency spectrum, aligning the discretized spatial frequency domain with the discrete pixelated spatial domain.

In Fourier optics, the non-negative, real intensity $I(x)$ of a light wave at a point x is proportional to the square of the electric field amplitude $E(x)$ at that point in the spatial domain. The electric field itself is generally a complex quantity. Therefore, the far-field intensity after free-space propagation is

$$\begin{aligned}
I(x) &= |E(x)|^2 \\
&= E^*(x)E(x) \\
&= \sum_j \xi_j^* \sigma_j^* \delta_W^*(x - x_j) e^{-2iWjx} \sum_h \xi_h \sigma_h \delta_W(x - x_h) e^{2iWhx} \\
&= \sum_j \sum_h \xi_j \xi_h \sigma_j \sigma_h \delta_W(x - x_j) \delta_W^*(x - x_h) e^{2iW(h-j)x} \\
&= \sum_{jh} \xi_j \xi_h \sigma_j \sigma_h \delta_W^2(x) e^{2iW(h-j)x}
\end{aligned} \tag{12}$$

Since δ_W , ξ_j and ξ_h are real numbers, their complex conjugates equal themselves. The double sum represents the interaction between points j and h on the SLM, contributing to the overall intensity. $e^{2iW(h-j)x}$ accounts for the phase difference between these points related to spatial frequencies. Since the delta functions $\delta_W(x - x_j)$ and $\delta_W(x - x_h)$ are localized around x_j and x_h respectively, the product is zero unless $x_j = x_h$. Therefore, $\delta_W(x - x_j) \delta_W(x - x_h) = \delta_W^2(x)$ is implemented when it is significant to consider the integration over the spatial overlap. It is the squared function of the spatial distribution, which approximates the effect of the spatial overlap of the delta functions.

The intensity of the light at each pixel corresponds to the spin state. A higher (lower) intensity for the spin value of +1 (-1). Spin-spin interaction refer to how the spin at one point (i.e., the phase of the light wave at that point) affects that of another, which is induced by acting on the intensity $I(x)$ on the detection plane after free-space propagation. To adjust the phase σ_j on the SLM, the observed intensity pattern $I(x)$ needs to match the target pattern $I_T(x)$. This is achieved by minimizing their difference, cost function $f = ||I_T(x) - I(x)||$, which corresponds to minimizing an effective Hamiltonian H .²

$I(x)$ is defined as the amount of light energy per unit area. When we integrate $I(x)$ over a spatial domain, we sum up the total intensity (or energy density) across the region. The integral can be thought of as related to the total energy of the intensity pattern, as a measure of the overall strength of the intensity pattern.

²In physics, minimizing a certain function is often equivalent to minimizing the energy of a system, described by a Hamiltonian H . In this case, the process of adjusting the phases to match the target intensity can be viewed as minimizing an effective Hamiltonian.

$$\mathbf{Energy}_{\text{total}} = \int I(x)dx \quad (13)$$

However, in the context of measurement, the energy is expressed as

$$\mathbf{Energy} = \int [I(x)]^2 dx \quad (14)$$

which provides a measure of the uniformity and capture how well both patterns across the entire spatial domain. Squaring the intensity emphasizes regions of higher intensity more than those of lower intensity to ensure the differences in areas of higher intensity are given more weight in the comparison. The use of square in the normalization condition helps ensure the distribution and consistency of the pattern.³

Since both the resulting intensity pattern $I(x)$ and the target intensity pattern $I_T(x)$ approximately equal, meaning that both their distribution and magnitude across the spatial domain, i.e., the integral for their squares over the spatial domain are very similar after normalization (both have been adjusted or scaled so that their power are comparable)

$$\int [I_T(x)]^2 dx \simeq \int [I(x)]^2 dx \quad (15)$$

In Ising model, the Hamiltonian H represents the energy of a system of interacting spins σ_j and σ_h . The minus sign indicates that the energy is minimized for a more stable system when spins are aligned in particular way. The coupling constant J_{jh} expresses the geometry of the spin lattice, strength and the electromagnetic (ferromagnetic or anti-ferromagnetic) nature between spins at positions j and h . Taking the effective forms of Ising Hamiltonian⁴

$$H = - \sum_{jh} J_{jh} \sigma_j \sigma_h \quad (16)$$

, which corresponds to the physical process of matching the observed and the target intensity pattern by adjusting the phase ϕ_j correlating to the spin σ_j on the SLM.

The spin interactions J_{jh} , i.e., the coupling strength between these spins can be influenced by SLM in how the phases at different points contribute to

³In statistical mechanics or signal processing, the square of a quantity is often related to variance or fluctuation. The reason for the squared values are often used when comparing distribution or when normalizing function to ensure both the magnitude and distribution match.

⁴Another form is the modified Ising Hamiltonian is $H = - \sum_{jh} J_{jh} \sigma_j \sigma_h - \sum_j h_j \sigma_j$, where h_j is an external magnetic field acting on spin j . This term represents a uniform external magnetic field that acts on all spins with equal strength. It can be used to introduce a bias towards a particular spin orientation, affecting the equilibrium state of the system. The modified Hamiltonian allows for more flexibility in modeling for studying a wider range of phenomena and applications.

the overall far-field intensity. The integral accounts for the contribution to the interaction between point j (with amplitude ξ_j and phase ϕ_j relating to spin σ_j) and point h (with amplitude ξ_h and phase ϕ_h relating to spin σ_h) due to the overlap of their electric fields ($E(x)^*$ and $E(x)$) when propagated and modulated by the target intensity pattern. It is given by

$$J_{jh} = 2\xi_j\xi_h \int I_T(x)\delta_W^2(x)e^{2iW(h-j)x}dx \quad (17)$$

couplings reduce to

$$J_{jh} = 2\pi\xi_j\xi_h\tilde{I}_T[2W(j-h)] \quad (18)$$

which indicates that the interaction matrix is set by the input amplitude modulation along with the Fourier transform of the far-field target image.

2.2 Spin Configuration and Intensity Pattern

When the spin configuration is encoded into the light wave, it produces a particular intensity pattern on the detector by setting the phase of light wave. Consider a 4×4 grid of spins, the configuration is as follow:

$$\begin{bmatrix} 0 & 0 & \pi & 0 \\ \pi & 0 & 0 & \pi \\ 0 & \pi & \pi & 0 \\ \pi & 0 & 0 & \pi \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & -1 & 1 \\ -1 & 1 & 1 & -1 \\ 1 & -1 & -1 & 1 \\ -1 & 1 & 1 & -1 \end{bmatrix}$$

The phase-modulated light waves from different pixels interfere with each other, creating a specific intensity pattern of a checkerboard-like structure on the detector. Regions with similar phases would interfere constructively, leading to higher intensity, while regions with different phases would interfere destructively, leading to lower intensity. If the specific intensity pattern on the detector matches the desired target intensity, then this spin configuration would be considered as the solution to the Ising problem.

2.3 Near-Field and Far-Field

The near-field is the region of space immediately adjacent to the SLM. In this region, the light intensity pattern is strongly influenced by the specific phase modulation introduced by the SLM. The near-field pattern can be quite complex and may exhibit rapid variations in intensity due to the localized nature of the phase modulation.

The far-field is the region of space at a significant distance from the SLM. In this region, the light wave has had time to propagate and spread out. The

far-field intensity distribution is essentially the Fourier transform of the near-field phase distribution. This means that the spatial frequencies present in the near-field pattern are mapped onto the spatial frequencies in the far-field. The far-field pattern is often simpler and more uniform than the near-field pattern, as the high-frequency components of the light wave tend to be attenuated over distance. The far-field constraint ensures that the generated intensity pattern matches the desired target intensity. This is equivalent to ensuring that the Fourier transform of the SLM phase distribution is consistent with the target.

2.4 Intensity to Long-Range Interaction

The interaction passes from short-range to long-range by changing the spatial profile of $I_T(x)$. When the spatial profile of $I_T(x)$ is localized to a bright concentrated spot, the interactions between spins are primarily confined to nearby neighbours, which is similar to nearest-neighbour interaction in lattice model. However, when its profile become more extended, the interaction between spins can reach farther distances, such that the spins can influence each other over larger separations. Therefore, spatial profile of $I_T(x)$ can be treated as a knob to control the range of interactions in the system for studying different types of spin models and exploring their properties, such as phase transitions.

To achieve longer-range interactions, broader function like using larger values for the width parameter of the Gaussian distribution can be employed to introduce a degree of overlap between the spatial distribution between different spins, leading to interactions that extend beyond nearest neighbors. Besides, modifying the phase gradient across the system can influence the phase between the lights fields from different spins, extending the interactions over distances. Also, adjusting the complex amplitude of both spins can create interference patterns that favor interaction in the long-range distance.

By adjusting the amplitude mask, the strength of the interaction between different pairs of spins can be altered to implement Mattis model, which are a class of spin-glass models with specific interaction patterns.

$$\frac{J_{jh}}{jh} \quad (19)$$

The coupling is inversely proportional to the product of the spin indices.

The Ising machine then iteratively conduct the following procedure:

1. A spin configuration is generated by modulating the wavefront with binary phases on the SLM.
2. The resulting intensity distribution $I(x)$ is measured in the far-field. 3. The detected image $I(x)$ is compared with the target $I_T(x)$.

3. The difference between $I(x)$ and $I_T(x)$ is used as feedback to adjust the SLM phases.
4. The evolves towards minimizing the cost function $f = |I(x) - I_T(x)|$, which corresponds to finding the ground state of the Ising model.

2.5 Holographic Nature

The holographic nature of the detected image in SLM is due to the interference patterns by phase modulation that arise from the interaction of multiple light waves, where the information about the original object (spin configuration) is encoded in the intensity distribution of the light field. The detected far-field image have holographic fringes due to the diffraction pattern from the Fourier transform of the near-field distribution as the light waves propagate over a long distance.

2.6 Resolution

Holoeye LC-R 720 is a specific model of SLM that uses liquid crystals to control the phase of the light wave. 1280×768 Pixels indicates the resolution or number of individual pixels on the SLM. $20 \times 20 \mu\text{m}$ is the size of each pixel on the SLM. $N = L \times L$ Spins suggests that the SLM is used to represent a 2D lattice of N spins, where L is the linear dimension of the lattice. CCD Camera is a detector that captures the intensity distribution of the light. The camera pixels are grouped into 16×16 blocks, resulting in $M = 18 \times 18$ spatial modes. This is likely done to reduce the impact of speckle noise. A single spatial mode can output a focused spot of light. The uniform amplitude associate with the spins are assumed to be constant. The experiment conduct random initialization of SLM phases to create a speckle pattern in the far-field. The laser's wavelength and the focusing optics determine the spatial resolution of the system, of which affects the minimum size of the pixels that can be effectively controlled on the SLM.

2.7 Pixel-to-Spin Mapping

Consider a grid of pixels on the SLM, each pixel can be thought of as a small window through which the light wave passes. By controlling the phase of the light wave within each window, the SLM effectively creates a pattern of phase variations across the wavefront. This pattern corresponds to the specific spin configuration being represented.

2.8 Monte-Carlo Algorithm

Afterwards, a Monte-Carlo method is implemented such that a small cluster of spin is randomly flipped instead of updating individual spins to improve the

efficiency of the algorithm. After updating a cluster, the far-field intensity distribution is measured, then the change in spin configuration is accepted only if it reduces the difference between the measured intensity and the target image. This approach is similar to the Metropolis-Hastings algorithm. The Metropolis-Hastings algorithm is a Markov Chain Monte Carlo (MCMC) Method used to generate samples from a probability distribution (which is often the Boltzmann distribution) that may be difficult to sample directly, which is particularly useful when the normalization constant of the distribution is unknown. By generating these samples, the properties of the distribution can be estimated, such as the mean, variance, the probability of certain events.

Steps for implementing the algorithm:

1. Initializing the state
2. Proposing a new state based on a proposal distribution
3. randomly decide whether to accept or reject the proposed state based on the acceptance probability
4. If the proposed state is accepted, update the current state to the new one. Otherwise, keep the current state.
5. Repeat steps 2-5 for a specific number of iterations.

The experiment does not require explicit knowledge of the Hamiltonian that may directly influence the spin evolution. Whereas, the target image and the spin configuration act as an implicit representation of the Hamiltonian by minimizing the cost function $f = ||I_T(x) - I(x)||$ that effectively minimizes the energy associated with the Hamiltonian. Since the measurement and the feedback loop can couple the phase ϕ relating to the spin σ on the SLM plane to the detected intensity, spin-spin interactions are induced by manipulating the intensity on the detection plane. This approach provides more flexibility to explore a wider range of problems and models.

2.9 Magnetization

The magnetization of the system can be used as a physical observable to track the evolution of the system, that a change in magnetization indicates a change in the overall spin configuration. Magnetization can be interpreted as a measure of the overall polarization or coherence of the light wave. Also, it can identify phase transition. When there is spontaneous magnetization in the system, a majority of spins are aligned in the same direction. The system exhibits a net magnetization ($m > 0$), which corresponds to ferromagnetic state below critical temperature T_c . While the spins are randomly oriented above T_c , there is low or no net magnetization then the phase transition is in paramagnetic state.

The system is disordered and exhibits no long-range order.

If there is in the ferromagnetic domains, it indicates a large degree of order in the connected regions of aligned spins. The almost equal probability distribution of spins with $m < 0$ and $m > 0$ suggests a balanced system without a net magnetization, expected in the absence of external magnetic fields in the system. The sizes of the ferromagnetic domains can be varied due to the imperfect uniformity of the system caused by fluctuation in the local interactions or external condition. The system can also experience opposite magnetization phase that all the spins are aligned in the opposite direction.

In general, a high magnetization often corresponds to a low-energy state. This is because a well-ordered system with a large number of aligned spins tends to have a lower energy than a disordered system. The ground state of the Ising model is typically characterized by a high magnetization. Therefore, a high fidelity (i.e., a high probability of finding the ground state) often correlates with a high magnetization.

By calculating the correlation length, the state can be determined. A long correlation length is associated with a ferromagnetic state.

2.10 Spin Alignment

In the photonic Ising machine, the phase transition is related to the alignment of spins. At high temperatures (analogous to a disordered state), the spins are randomly oriented. As the temperature decreases (analogous to cooling), the spins may become aligned in a more ordered state (e.g., ferromagnetic). A sudden change in the correlation length as the temperature is varied can indicate a phase transition from a disordered state to an ordered state.

2.11 Random Phase Fluctuation (RPF)

RPF can be interpreted as a form of noise or thermal energy that affects the system's behavior. The level of noise can be related to the effective temperature of the system. Higher noise levels correspond to higher effective temperatures. By controlling the level of noise, one can effectively manipulate the system's behavior and the rate at which it converges to a low-energy state. However, these phase variants of light wave are uncontrollable arising from the noise in the system.

RPF can be extracted by directly measured the fluctuations in the intensity of the light detected by the CCD camera, and then by calculating the standard deviation or variance of the intensity fluctuation over time. By using the

interferometry techniques and analyzing the correlation between the phase fluctuation at different points in the system, this information can also be gained. Measuring the spin configuration multiple times with analysis on the variability between measurements and calculating the entropy of the spin configuration distribution to quantify the level of disorder or randomness can help obtaining the fluctuation. The level of noise can be inferred from the system's sensitivity to the external perturbation.

2.12 Simulated Annealing

Simulated Annealing (SA) is a classical probabilistic optimization algorithm inspired by annealing process in metallurgy, which involves gradually reducing a temperature parameter, similar to cooling a material.

Steps for implementing the algorithm:

1. Initialization: Start with a high temperature and a random initial state.
2. Perturbation: Propose a new state by perturbing the current state.
3. Acceptance: If the new state has lower energy, accept it unconditionally.
4. If the new state has higher energy: Accept it with probability $e^{-\frac{\Delta E}{T}}$, where ΔE is the energy difference and T is the temperature.
5. Cooling: Gradually reduce T according to a cooling schedule.
6. Repeat: Repeat steps 2-4 until a satisfactory solution is found or a maximum number of iterations is reached.

Initially, SA explore a wide range of configuration at a higher temperature, and then it becomes more selective and focus on lower-energy state as the temperature decreases. SA gradually increase the clusters helps better opportunity to find deeper energy minima and avoid get trapped in shallow local minima. The dependence of the system's temperature is a core principle of SA to control the acceptance probability of moves, which is determined by random phase fluctuations. The gradual increase in magnetization $|m|$ (gradual alignment of the spin) and the monotonic decrease in Hamiltonian H (moving toward a lower-energy state) are consistent with the behavior of SA to minimize the energy of the system and to settle the system in a more ordered state.

2.13 Max-Cut Problem

SA can be applied to solve Max-Cut problem for its effectiveness in handling Combinatorial Optimization (CO) problems with discrete variables and complex energy landscapes. Max Cut problem is a CO problem that involves partitioning a graph into two sets such that the number of edges connecting vertices in

different sets is maximized. This is an NP-hard problem that it is computationally difficult to find the optimal solution for large instances. SA can explore the solution and avoid getting trapped in local minima by gradually cooling the system, which is a common practice in solving Max Cut problem.

2.14 QAOA

QAOA is a hybrid quantum-classical algorithm that leverages both classical and quantum computing resources, which can be applied to various CO problems. However, for large-scale Ising problems, the quantum resources required for QAOA might become prohibitive. Also, there may be hardware constraints that affects the availability and performance in choosing QAOA as the computing algorithm, the choice of its parameters as well.

2.15 Möbius ladder graph

Möbius ladder graphs are often used as examples or test cases in graph theory and algorithms. In particular, an unweighted (all edges in the graph have the same weight) Möbius ladder graph is a specific type of graph constructed from a cycle graph (each vertex is connected to its two neighbors) by adding a set of edges that connect opposite vertices of the cycle, forming a ladder-like structure with a twist (one of the edges connecting opposite vertices of the cycle is crossed over the other edges). It happens when all amplitudes are equal ($\xi_j = \xi_h = \xi_0$) i.e., each spin has the same overall intensity. It is non-planar, that it cannot be drawn on a plane without edge-crossing. It has a Hamiltonian cycle that visits every vertex exactly once. The rotational symmetry allows the graph can be rotated by certain angles without changing its structure.

In the context of photonic Ising machine, the spins represent the vertices of the Möbius-Ladder graph, and the edges between vertices represent the couplings between spins that determine the strength and nature of the interactions between them. The graph structure refers to a particular graph topology where the spins are arranged in a circular pattern with additional connections between opposite vertices, forming a ladder-like structure with a twist. A fully-connected vertices structure means that every spin is connected to every other spin regardless of their distance in the graph. The spin configuration in the Möbius-Ladder graph corresponds to the target intensity $I_T(x)$ when the system reaches its equilibrium state.

Since focus is on a single spatial mode exhibiting a bright localized spot, which shows that the graph is a circular pattern. Although the base of the graph is a cycle, while the spins can be both on the rim and within the interior of the circle. The circular pattern is projected onto the 16×16 CCD camera pixels, which represent different spatial modes for each corresponding to a specific

pattern of light intensity throughout the distribution. The projection of the spin

2.16 Energy Probability Distribution Function (PDF)

Evaluating PDF is to assess the distribution of energies among all possible spin configuration that satisfies the far-field constraint of producing the target intensity, which helps to determine how likely it is for the machine to find a low-energy solution.

2.17 Quantized Phase-Retrieval (QPR) Algorithm

QPR is used to generate Binary Phase Distribution (BPD) corresponding to particular spin configuration from $I_T(x)$. It iteratively adjusts the phase distribution to minimize the difference between $I(x)$ and $I_T(x)$.

2.18 Spontaneous Symmetry Breaking

Spontaneous symmetry breaking occurs when a system with a symmetric Hamiltonian (in this case, the Ising Hamiltonian) evolves into a state that lacks the original symmetry. In the absence of an external magnetic field, the Ising Hamiltonian is symmetric under the transformation of all spins flipping their values. However, the system may spontaneously break this symmetry by adopting a state with a net magnetization

2.19 Mean Field Approximation (MFA)

The mean-field solution is an approximate method that simplifies the interaction between spins for solving the Ising model. It assumes that each spin interacts with an average field created by all other spins. By comparing the observed spin configurations to the mean-field solution, the effective temperature of the system can be estimated. This is because the mean-field solution depends on the temperature. At the critical temperature (T_c), the mean-field approximation breaks down. This means that the equations used to describe the system become divergent or invalid.

MFA is used to provide an initial guess for the system's configuration. Then, a Monte Carlo method, such as Metropolis-Hastings or Simulated Annealing (SA), is used to refine the solution and explore the configuration space more thoroughly. The mean-field approximation can help to accelerate the convergence of the Monte Carlo algorithm, especially for large systems. Also, it simplifies the calculation of the system's energy and other properties, making it computationally more efficient.

2.20 Experimental Setup

Nematic liquid Crystal Reflective Modulator (SLM) controls the phase of the light wave for representing the spins states in the Ising machine. Its resolution is 1280×768 pixels, which can control the phase of the light wave at 1280×768 points. The pixel pitch of $20 \times 20 \mu\text{m}$ determines the spatial resolution of the SLM. Smaller pixel pitch allows for finer control of the light wave's phase. The active area is a rectangular aperture interacting with light that defines the region of the SLM and determines the total number of spins ($N = L \times L$) that can be represented. The low residual intensity phase modulation is crucial for accurately representing the spin states without introducing unwanted effects. The filtered Light by spatial filtering is being focused onto the CCD camera by focusing lens, which measures the intensity distribution of the light to infer the spin configuration. The grouping 16×16 blocks to form 18×18 spatial mode helps to average out noise and improve signal-to-noise ratio.

The higher the resolution of the SLM for smaller pixel pitch, the higher quality of the overall optical setup, the higher resolution of the CCD camera by having higher pixel density, the finer control of the phase with more detailed spin configuration and the more accurate of the measurement.

Using a lens with higher numerical aperture can focus the light beam more tightly, which reduces the diffraction effects and improve resolution. A cleaner and more focused light beam can be created by minimizing optical aberrations.⁵ Precise alignment can also minimize errors and improve resolution.

2.21 Interpretation of the Result

The majority of the spins in the system are pointing in the same direction indicating a ferromagnetic-like state, corresponds to a low-energy configuration that minimize the total energy due to their interactions. The monotonically decrease Hamiltonian displays the system is moving towards a lower-energy state, which is the desired outcome in optimizing problems. The plateau in the Hamiltonian shows that the system has reached a stable, ordered state where a further change in the spin configuration would require a significant increase in energy.

The 100 Phase-retrieved spin configurations provide a set of potential solutions that satisfy the far-field constraints. By evaluating the energy of these solutions, we can obtain a distribution of energies.

87% probability of the correct minimum solution with 5 % lowest energy refer to 87 out 100 trails of successfully identified the spin configuration that corresponds to the global minimum of the Hamiltonian, and be able to find the solutions that lie within the top 5% of the lowest-energy configuration. It is independent of the way the ground state has been found, which means that even if a different algorithm or optimization technique were used to solve the

⁵spherical aberration, chromatic aberration

Ising problem, the machine would still be expected to find the correct minimum solution with approximately 87% probability. Therefore, it implies that the machine is efficiently robust and can consistently find reliable, accurate solutions to the Ising problem, regardless of the optimization approach used.

In Figure 2(e), there are ferromagnetic domains and opposite magnetization phase being observed. The multiple images arrangement in grid may suggest the evolution of spin configuration as the system progress. The white and blue regions represent spin configuration with values of +1 and -1 respectively. The distribution of these regions reveals the ferromagnetic domains with different magnetization. The sub-figure on the left depicts the large size of white region with $m > 0$ while that on the right depicts the large size of blue region with $m < 0$. The small size of the ferromagnetic clusters indicate that multiple small regions where spin are aligned in the same direction, and are separated by regions with opposite magnetization. For the various sizes in the ferromagnetic domains, it is due to the imperfect uniformity caused by the fluctuation in the local interactions. The almost equal probability for the spin states with $m < 0$ and $m > 0$ suggests no preferential direction for the spins to align, constituting a balanced system.

The images shows the formation or disappearance of ferromagnetic domains, indicating the phase transition in the system. The evolution of the domain structure provide information in the interaction between spins and the system's overall behavior. The presence of ferromagnetic domains with opposite magnetization is consistent with spontaneous symmetry breaking⁶.

Mean Magnetization represents the average spin value in the system. A positive m indicates a net alignment of spins in one direction, while a negative m indicates alignment in the opposite direction. The equation describe the relationship between the mean magnetization and the temperature for a system undergoing a phase transition. Below is a simplified version of the mean-field approximation for the Ising model.

$$m = \tanh\left[\frac{T_c}{T}m\right] \quad (20)$$

2.22 Spin Spatial Auto-correlation

Spin Spatial Auto-correlation $g(r)$ is a function that measures the correlation between the spins at two points separated by a distance r . It provides information into the spatial structure of the spin system, such as the presence of domains and the range of interactions between spins.

⁶The system, initially symmetric, has spontaneously chosen a particular configuration that breaks the symmetry.

$$g(r) = e^{-\frac{r}{\xi}} \quad (21)$$

A high value of $g(r)$ indicates strong correlations between spins at a distance r that the spins at those points are more likely to be aligned in the same direction, while a low value indicates weak correlations that the spin alignment would not be aligned to each other or even opposite to each other. Correlation length ξ determines the rate at which the correlations decay with distance. A larger of its value indicates longer-range correlations that $g(r)$ decays slowly with distance. It is in the ferromagnetic state.

The equation can be used to determine the critical temperature T_c , and ξ provides an estimate of the average domain size in the system. Larger domains typically indicate a stronger degree of magnetic order. A large ξ at T_c indicates that the spins are strongly correlated over long distances.

Spin Spatial Correlation can also be express as follow:

$$g(r) = \langle \sigma(x)\sigma(x+r) \rangle \quad (22)$$

$\sigma(x)$ is the spin value at position x . $\langle \dots \rangle$ denotes the average over the entire system.

To measure $g(r)$, the spin configuration from the measured intensity distribution has to be determined, and then the $\sigma(x+r)$ for each distance r is calculated to obtain $\sigma(x+r)$ by repeating the process.

Both are the common model to describe the decay of the spin-spin correlation in a system.

ξ can be determined by fitting the equation to the measured auto-correlation function, which involves adjusting the value of ξ until the equation best matches the experimental data.

(Short-range interactions is a simplified model in consideration, since long-range interactions can significantly affect the behavior of the system.)

In the mean field approximation, ξ is expressed as follow

$$\xi = R_*(1 - \frac{T}{T_c})^{-\beta}, \quad \beta = \frac{1}{2} \quad (23)$$

The equation describes how the correlation length ξ varies with temperature. Critical exponential β characterizes the rate at which ξ diverges as the temperature approaches T_c . β provides information about the universality class of the system and can be used to compare the system's behavior to other systems with similar critical exponents. R_* is a minimum cluster length that sets a lower limit on the size of the clusters considered in the analysis. It helps to avoid considering fluctuations due to small-scale effects. It can influence the accuracy of the estimated critical temperature and the behavior of the system

near the phase transition.

The divergence of the correlation length at T_c indicates the system undergoes a phase transition at this temperature. The spins become strongly correlated over long distances, leading to a change in the system's behavior.

The fact that the observed ground states are consistent with a mean-field Ising model at a fixed temperature suggests that the mean-field approximation is a reasonable approximation for this system, even though it breaks down at the critical temperature.

The observed critical temperature ($T_c = 0.83 \pm 0.02$) indicates the temperature at which the system undergoes a phase transition. A low $\frac{T}{T_c}$ ratio suggests that the system is well below its critical temperature and is likely in a highly ordered state, such as a ferromagnetic state. In this regime, the spins have strong correlations over long distances, leading to a well-defined magnetization and domain structure. The value of $\frac{T}{T_c}$ might have been obtained by fitting the experimental data to a theoretical model, such as the mean-field approximation or a more accurate model for the system. The uncertainty of ± 0.02 indicates the precision of the measurement or the level of noise in the data. If $\frac{T}{T_c}$ were nearly 0, it would imply that the system is operating very close to absolute zero, which is an extreme condition that is often difficult to achieve experimentally. However, the observed value of $T_c = 0.83 \pm 0.02$ indicates that the system is operating at a significantly higher temperature, but still well below the critical temperature.

2.23 Phase Coherence in Light

In the context of light, phase coherence means that the light waves maintain a consistent phase relationship. This leads to interference effects and can result in a highly ordered or structured pattern of light intensity.

In the photonic Ising machine, the phase of the light wave is used to represent the spin state. Therefore, a system with strong spin correlations would also exhibit a high degree of phase coherence among the light waves representing those spins.

Photonic Ising machine can simulate a very large number of spins. The number of simulated spins can be controlled by adjusting the size of the active area on the SLM. The magnetization and the fidelity can be varied by altering the number of spins.

Figure 3(a) and Figure 3(b) demonstrates how $\overline{m}$ and fidelity of the system changes with the number of spins. It is found that the performance of the machine is insensitive to the number of spins, of which the interactions between

spins are uniform. However, due to the limitations in the spatial resolution of the detection plane, there is a slight decrease in $\overline{m}$ and fidelity for large N system.

7

Owing to finite-size effects, the small size of the system significantly influence the magnetization of the system.

8

However, as the system size increases, finite-size effects become less significant and the behavior becomes more consistent with the predictions of mean-field theory. In a larger system with more spins, the statistical fluctuations are typically smaller and a well-defined single decay pattern emerges in $g(r)$. This can lead to a more consistent and predictable behavior, including a higher average magnetization. This suggests that the domain structure is determined by intrinsic properties of the system, rather than being strongly influenced by the system size. A larger system can accommodate larger domains of aligned spins, which can contribute to a higher overall magnetization. Therefore, it suggests that the Ising photonic machine is highly scalable.

2.24 Mattis Spin Glass

Mattis spin-glasses are a specific type of spin glass model which can be implemented in the photonic Ising machine. Besides, amplitude modulation and phase modulation in SLM, the coupling between spins is determined by generating random coupling J_{jh} with both strong and weak interactions in terms of a set of random variables ξ_j and ξ_h representing the electric field amplitude of light waves.

According to the theoretical solution of the Mattis model, the ground state (lowest-energy configuration) of the system is closely related to the distribution of ξ_i . The expected ground state is either the same as the ξ_i configuration or its reversal, except for the weakly interacting regions where the spins are randomly oriented.

⁷A lower spatial resolution in the detection plane means that the camera cannot distinguish between individual spins that are too close together. This can lead to errors in the measurement of the spin configuration and affect the overall performance of the machine. Therefore, the limitations in spatial resolution might contribute to a slight decrease in magnetization and fidelity, especially for large systems where the spins are densely packed.

⁸When the system size is small, the effects of the boundary conditions can become more pronounced that ξ grows linearly with the system size. This can lead to deviations from the behavior of a larger system. Smaller systems tend to have fewer degrees of freedom, which can limit the range of possible configurations and reduce the fluctuations in the system's properties.

In Figure 4(a), the two top panels likely represent different configurations of the coupling matrix J_{jh} . The black and white squares might indicate strong and weak interactions, respectively. The bottom panel shows the corresponding spin configuration, with blue and white squares representing spins with different orientations. The black and white squares in the top panels likely represent different values of ξ_j . Black squares might correspond to strong interactions ($\xi_j = \xi_0$), while white squares might represent weak interactions ($\xi_j = 0$). The spin configuration in the bottom panel is influenced by these coupling strengths. In regions with strong interactions ($\xi_j = \xi_0$), the spins are more likely to be aligned in the same direction (ferromagnetic-like), while in regions with weak interactions ($\xi_j = 0$), the spins might be randomly oriented.

The terms "strongly" and "weakly" might be used in a relative sense. While all interactions are technically

2.25 Spin- ξ Correlation

C measures the correlation between the spin configuration σ_j and the random variable ξ_j that determine the coupling strength.

$$C = \sum_j \sigma_j \frac{\xi_j}{\xi_0}, C = \pm 1 \quad (24)$$

ξ_j is a normalization factor to ensure the dimensionless quantity of C. A high value of C indicates a strong correlation between the spin configuration and the expected ground state based on the ξ_j values, while a low value of C suggests that the measured spin configuration deviates from the expected ground state.

2.26 Question

Scalability

What are the limitations on the number of spins that can be simulated using this approach? How does the performance scale with the system size? The paper discusses the dependence of the machine's performance on the number of spins, which can provide insights into its scalability.

Noise and Error

How do noise and errors in the system affect the accuracy of the results? What techniques can be used to mitigate these effects? The paper might have analyzed the effects of noise and errors in the system, and discussed potential mitigation strategies.

Comparison with Other Platforms

How does this photonic Ising machine compare to other implementations, such as superconducting qubit-based or trapped-ion-based systems? The paper might have compared the performance of the photonic Ising machine to other implementations, or discussed the unique advantages and disadvantages of the pho-

tonic approach.

Applications

What are the potential applications of this photonic Ising machine beyond the examples discussed in the paper? The paper might have explored potential applications of the machine, such as solving optimization problems or simulating quantum systems.

Mattis Model

How does the performance of the machine vary for different types of spin-glass models, such as non-Mattis models?

Hamiltonian Optimization

Can the machine be used to solve other types of optimization problems beyond the Ising model?

Quantum Advantage

Is there a potential for this photonic Ising machine to demonstrate a quantum advantage over classical computers for certain problem instances?

Experimental Challenges

What are the main experimental challenges in implementing and operating this photonic Ising machine? How can these challenges be addressed?

2.27 Conclusion

It demonstrates the potential of photonic platforms for implementing large-scale Ising machines, which are crucial for solving complex optimization problems. Photonic systems can potentially be scaled to a much larger number of qubits (or spins, in this case) compared to other quantum computing technologies. The use of spatial light modulation to represent and manipulate spin states is a novel approach that offers unique advantages, such as flexibility and scalability in terms of interactions and Hamiltonian that can be implemented. The paper highlights the potential of photonic platforms for quantum computing, as Ising machines are closely related to quantum annealing and other quantum optimization algorithms. The paper provides a benchmark for the performance of photonic Ising machines on various Ising problems, including their ability to solve challenging problems and their limitation. It highlights the capabilities of classical systems and identify areas where quantum algorithms might offer a significant advantage. Photonic Ising machines can be used as a component in hybrid quantum-classical systems, where classical computers can be used to preprocess or post-process the data, while quantum algorithms handle the computationally challenging parts. The results presented in the paper could have implications for various applications, such as materials science, machine learning, and optimization problems in engineering and finance.

However, photonic systems can be susceptible to noise and errors, which can degrade the performance of the Ising machine. Developing techniques to mitigate these effects is crucial. Implementing complex Hamiltonians or large-scale systems can be challenging in photonic platforms. Advances in optical components and control techniques are needed to address these challenges. Demonstrating a clear quantum advantage over classical computers for solving Ising problems using photonic platforms remains an active area of research.

3 Terminology

Active Area

The rectangular aperture that defines the region of the SLM that is used to interact with the light.

All-to-All Interactions

Every spin interacts with every other spin, regardless of their distance. This is equivalent to having a coupling matrix J where all elements J_{jh} are constant.

Amplitude Modulation

The amplitude of the light wave is controlled or varied, which is crucial for encoding information or manipulating the spin states.

Beam Expansion

The laser beam is expanded to increase its size and intensity before it reaches the SLM.

Computational units

Each pair of pulses can represent a computational unit or "spin" in an Ising model. The phase or amplitude of these pulses can be manipulated to represent the state of the spin (+1 or -1).

Continuous Wave Laser

The laser emits light continuously, as opposed to pulsed lasers.

Correlation Length This is a measure of the spatial extent over which spins are correlated. A larger correlation length indicates that spins are more likely to be aligned with each other over longer distances.

Ferromagnetic Domain

Ferromagnetic domains are regions within a material where the magnetic moments of the atoms are aligned in the same direction. In the context of the photonic Ising machine, these domains correspond to regions where the spins are aligned in the same direction.

Fidelity

This is a measure of how accurately the machine finds the ground state (the lowest-energy configuration) of the Ising model. A high fidelity indicates that the machine is able to consistently find the correct solution.

Frequency Domain

The effect of this rectangular function (finite pixel size) in the spatial domain is represented by a sinc function in the frequency domain, which is what $\tilde{\delta}_W(k-k_j)$ models.

High speed

Light travels at high speed, enabling fast computations.

Ising model

It is a simplified model of magnetism where particles (spins) can be in one of two states (+1 or -1). Interactions between these spins determine the overall energy of the system. Finding the configuration with the lowest energy (the ground state) is a fundamental problem in statistical physics. Interestingly, many complex optimization problems can be mapped onto the Ising model. By representing problem variables as spins and defining appropriate interactions, finding the optimal solution becomes equivalent to finding the ground state of the Ising model. This approach has gained significant attention due to its potential to leverage insights from physics for computational challenges.

Large-Scale Implementation

The research likely focuses on scaling up the photonic Ising machine to handle larger problems, potentially involving a large number of spins or interacting elements.

Magnetization

In a system of spins, magnetization is a measure of the net alignment of the spins. It is calculated as the average spin value over all spins.

Mattis Models

These are a class of spin-glass models where the pairwise interaction can be expressed as the product of two independent variables. In other words, the coupling between spins j and h can be written as $J_{jh} = A_j * B_h$

Nanophotonic circuits

These integrate optical components on a chip to directly implement small-scale spin systems.

Opposite Magnetization Phase

This is a region where the spins are aligned in the opposite direction compared to the ferromagnetic domains.

Optical computing

The field of using light for computational tasks.

Optical parametric oscillators (OPOs)

Devices that generate multiple optical frequencies from a single input laser. It is a nonlinear optical device that converts an input laser beam into two output beams with lower frequencies (idler and signal).

Optimization

Such machines are designed to find ground states of Ising models, which correspond to the optimal solutions of certain types of optimization problems.

Parallelism

Optical systems can process multiple pieces of information simultaneously.

Phase Modulation

The SLM is configured to primarily modulate the phase of the light wave, rather than its amplitude.

Phase Transitions

The behavior of the system can be different under the influence of an external field, including changes in the location of phase transitions.

Photonic platforms

Systems that use light to perform computations.

Photonic Ising Machine

It is a type of computing device that leverages the principles of the Ising model for solving optimization problems. The Ising model is used to simulate interactions in systems with binary states, and it's particularly well-suited for addressing combinatorial optimization problems. While these systems offer the potential for high-speed computation due to the speed of light, several challenges persist:

- Controlling a large number of spins: Creating and manipulating a vast number of optical elements to represent spins is technically demanding.
- Precise interaction engineering: Designing optical systems to accurately mimic the desired spin interactions is complex.
- Measuring the ground state: Accurately determining the system's lowest energy state from optical measurements can be challenging.

Pixel $N \times M$ pixels indicates the resolution or number of individual pixels on the SLM.

Pixel Pitch

$N \times N \mu\text{m}$ is the size of each pixel on the SLM.

Pixel Spacing

The distance between the centers of adjacent pixels is $2W$.

Point-Like Target

This refers to a target image where the light intensity is concentrated at a single point.

Programmable Amplitude Mask

This is a device that can control the amplitude of the light wave at different

points.

Residual Intensity Modulation

Despite efforts to minimize it, there might be some residual variation in the light intensity due to imperfect phase modulation.

Random Initialization

The SLM phases are initially set to random values, creating a speckle pattern in the far-field.

Random Phase Fluctuation

Random phase fluctuations arise due to various sources of noise in the experimental setup, such as quantization noise from the CCD detector and imperfections in the spatial phase modulation.

Scalability

Some photonic platforms, like spatial light modulation, can potentially handle large datasets.

Spatial Filtering

The light is filtered to remove any unwanted components or noise.

Domain

The physical size of the pixel is described by a rectangular function.

Spatial Frequency

Spatial frequency represents the number of cycles of a spatial pattern per unit length.

Spatial Light Modulation

This refers to the use of devices like Spatial Light Modulators (SLMs) to control the amplitude, phase, or polarization of light beams in space. In the context of an Ising machine, SLMs is used to imprint the desired phase pattern onto the light wave, effectively controlling the spin configuration. SLM is another approach for optical computing, where the shape of light fields can represent data and perform computations. However, its application to Ising spin dynamics has been relatively unexplored prior to the discussed work.

Spatial Mode

A spatial mode is a specific pattern or distribution of light intensity within a beam. It describes how the light is distributed across the transverse (cross-sectional) plane of the beam. It defines the variation of light intensity across the beam's profile. The mode can have various shapes, such as Gaussian (bell-shaped), Hermite-Gaussian (multiple peaks and valleys), or Laguerre-Gaussian (donut-shaped). Different spatial modes are often orthogonal, meaning they do not interfere with each other. Spatial mode describe the quality of a laser beam, such as its divergence and ability to focused. Understanding it is crucial for de-

signing and analyzing optical systems. In the context of SLM, the interaction between the spins and their spin states can be known by measuring the intensity distribution of the light in different spatial mode.

Spatial Profile

This refers to the distribution of light intensity within the target image.

Speckle Noise

This refers to random fluctuations in the intensity of the light due to spatial phase variations. Averaging over multiple pixels helps to reduce the effects of speckle noise.

Spontaneous Magnetization

Even without an external magnetic field, a system can exhibit spontaneous magnetization, especially at low temperatures. This is due to the interactions between the spins, which can cause them to align cooperatively.

Spontaneous Symmetry Breaking

It occurs when a system with a symmetric Hamiltonian (the Ising Hamiltonian) evolves into a state that lacks the original symmetry due to thermal fluctuation or other factors. This means that the spins will no longer be randomly oriented but will align in a particular direction. For example, in the absence of an external magnetic field, the Ising Hamiltonian is symmetric under the transformation of all spins flipping their values. However, the system may spontaneously break this symmetry by adopting a state with a net magnetization.

Symmetric Hamiltonian

The Ising model, which describes the interactions between spins in the photonic Ising machine, has a symmetric Hamiltonian. This means that the system's energy is invariant under certain transformations, such as flipping all spins.

Target Image

This is the desired pattern or configuration that the machine aims to achieve. It's projected onto the photonic system.

Time-multiplexed optical parametric oscillators (TM-OPOs)

These generate multiple optical pulses that can represent the spins in an Ising model. They are a technique where multiple optical pulses are generated sequentially in time within a single optical parametric oscillator (OPO) cavity. Each of these pulses can be independently manipulated and represents a potential computational unit or "spin" in the context of Ising machines. TM-OPOs have been used to implement Ising machines, where the phase of each pulse represents the spin state. The interactions between the pulses can be engineered to mimic the interactions between spins in an Ising model.